

LARYAK, Russian Federation

WMO: 238670

Lat: 61.100N

Lon: 80.250E

Elev: 57

StdP: 100.65

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.0	(c) -38.1	(d) -44.7	(e) 0.0	(f) -41.7	(g) -40.8	(h) 0.1	(i) -37.8	(j) 6.6	(k) -15.5	(l) 6.2	(m) -14.7	(n) 1.0	(o) 240	(p) 0.636

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	29.2	18.9	27.5	18.4	25.4	17.5	20.3	26.6	19.2	24.9	18.3	23.7	2.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.2	22.7	17.2	12.4	21.6	16.1	11.5	20.9	58.8	26.3	55.1	25.1	52.0	23.6	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.0	(b) 6.1	(c) 5.3	(e) -43.4	(f) 31.4	(g) 3.6	(h) 1.6	(i) -46.0	(j) 32.6	(k) -48.1	(l) 33.5	(m) -50.2	(n) 34.3	(o) -52.8	(p) 35.5		
(a) 7.0	(b) 6.1	(c) 5.3	(e) -43.2	(f) 21.4	(g) 3.3	(h) 1.4	(i) -45.6	(j) 22.4	(k) -47.6	(l) 23.3	(m) -49.5	(n) 24.0	(o) -51.9	(p) 25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.6	-22.5	-18.4	-8.3	-0.3	5.6	17.0	17.8	14.2	8.7	0.1	-14.6	-19.6	(6)
(7)		DBStd	15.71	8.51	8.94	6.89	5.95	6.05	5.16	4.04	3.59	4.62	5.93	9.02	10.33	(7)
(8)		HDD10.0	4907	1009	796	567	311	168	6	0	5	79	310	739	917	(8)
(9)		HDD18.3	7378	1267	1029	825	560	397	86	61	135	289	564	989	1175	(9)
(10)		CDD10.0	669	0	0	0	2	31	216	241	136	39	3	0	0	(10)
(11)		CDD18.3	99	0	0	0	0	2	46	43	8	1	0	0	0	(11)
(12)		CDH23.3	897	0	0	0	0	23	441	362	70	2	0	0	0	(12)
(13)		CDH26.7	234	0	0	0	0	5	132	89	9	0	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.0	2.2	2.7	2.8	3.0	2.3	2.0	2.2	2.6	2.3	2.1	(14)	
(15)	Precipitation	PrecAvg	575	27	24	29	34	48	64	73	78	56	58	46	38	(15)
(16)		PrecMax	758	69	43	58	70	132	145	152	168	117	111	89	74	(16)
(17)		PrecMin	393	4	5	4	2	13	8	17	3	7	29	23	17	(17)
(18)		PrecStd	108	13	10	13	18	24	28	37	37	25	19	16	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.1	1.2	5.7	16.8	27.0	31.6	31.4	27.5	22.9	16.5	3.9	-0.4	(19)
(20)			MCWB	-4.6	0.1	1.3	9.3	15.5	19.9	21.1	18.6	15.5	10.6	3.2	-1.1	(20)
(21)		2%	DB	-6.2	-1.7	3.7	12.4	21.5	29.8	28.9	25.2	20.3	12.6	1.4	-2.0	(21)
(22)			MCWB	-6.8	-2.3	0.3	6.8	13.1	18.6	19.2	17.1	14.2	8.9	0.7	-2.7	(22)
(23)		5%	DB	-8.4	-4.6	2.1	10.1	18.6	28.2	27.3	23.5	18.1	9.5	-0.2	-3.9	(23)
(24)			MCWB	-9.1	-5.4	-0.5	5.4	11.8	18.4	18.6	16.6	13.1	7.1	-0.8	-4.6	(24)
(25)		10%	DB	-10.4	-7.2	0.7	7.9	15.2	26.2	25.3	21.4	15.7	7.6	-1.8	-6.1	(25)
(26)			MCWB	-10.9	-7.9	-1.2	4.2	9.9	17.6	17.9	15.8	11.9	5.8	-2.5	-6.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.6	0.3	2.2	9.8	16.1	21.7	22.3	19.6	16.3	11.6	3.3	-1.0	(27)
(28)			MCDWB	-4.2	1.1	4.0	15.5	23.8	29.5	29.6	25.6	21.4	15.8	3.9	-0.5	(28)
(29)		2%	WB	-6.6	-2.2	0.8	7.5	13.6	20.2	20.6	18.5	14.8	9.3	0.9	-2.6	(29)
(30)			MCDWB	-6.1	-1.6	3.1	11.9	20.0	27.6	26.4	23.1	18.9	12.2	1.3	-2.0	(30)
(31)		5%	WB	-9.0	-5.5	-0.1	5.9	12.1	19.0	19.6	17.4	13.7	7.4	-0.9	-4.5	(31)
(32)			MCDWB	-8.4	-4.7	1.8	9.3	17.9	26.5	24.7	21.7	17.4	9.3	-0.2	-3.8	(32)
(33)		10%	WB	-11.0	-7.9	-1.3	4.6	10.2	18.1	18.6	16.4	12.2	5.8	-2.5	-6.7	(33)
(34)			MCDWB	-10.5	-7.2	0.7	7.5	14.7	24.6	23.1	20.5	15.3	7.3	-1.8	-6.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.2	9.4	9.3	8.6	9.5	9.3	8.8	7.4	5.2	7.1	7.1	(35)
(36)			MCDBR	8.4	6.6	8.3	10.4	12.3	13.0	12.2	12.4	11.0	7.9	4.8	7.9	(36)
(37)			MCWBR	8.0	6.0	6.3	6.8	6.5	5.0	4.7	5.6	5.9	5.3	4.2	7.6	(37)
(38)		5% WB	MCDBR	8.0	6.6	8.2	9.5	12.1	11.9	10.7	10.7	9.9	7.4	4.0	8.1	(38)
(39)			MCWBR	7.8	6.0	6.7	6.7	7.0	5.3	4.8	5.7	5.6	5.3	4.1	7.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.258	0.299	0.349	0.369	0.381	0.402	0.444	0.410	0.359	0.326	0.264	0.245	(40)	
(41)		taud	1.953	2.004	1.955	2.053	2.285	2.328	2.181	2.268	2.369	2.243	2.003	1.895	(41)	
(42)		Ebn at Noon	422	641	733	810	839	826	772	769	748	623	410	257	(42)	
(43)		Edh at Noon	45	82	122	134	116	114	129	109	82	65	42	29	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.39	1.29	2.84	4.60	5.05	5.60	5.07	3.58	2.22	0.99	0.42	0.20	(44)	
(45)		RadStd	0.04	0.11	0.17	0.45	0.51	0.41	0.57	0.47	0.28	0.13	0.09	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (2 neighbors)		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A

Nomenclature: See separate page